

Chapter 36. RV32/64G Instruction Set Listings

One goal of the RISC-V project is that it be used as a stable software development target. For this purpose, we define a combination of a base ISA (RV32I or RV64I) plus selected standard extensions (IMAFD, Zicsr, Zifencei) as a "general-purpose" ISA, and we use the abbreviation G for the IMAFDZicsr_Zifencei combination of instruction-set extensions. This chapter presents opcode maps and instruction-set listings for RV32G and RV64G.

Table 72. RISC-V base opcode map, $inst[1:0]=11$

$inst[4:2]$	$inst[6:5]$	000	001	010	011	100	101	110	111 (>32b)
00		LOAD	LOAD-FP	<i>custom-0</i>	MISC-MEM	OP-IMM	AUIPC	OP-IMM-32	<i>reserved</i>
01		STORE	STORE-FP	<i>custom-1</i>	AMO	OP	LUI	OP-32	<i>reserved</i>
10		MADD	MSUB	NMSUB	NMADD	OP-FP	OP-V	<i>custom-2</i>	<i>reserved</i>
11		BRANCH	JALR	<i>reserved</i>	JAL	SYSTEM	OP-VE	<i>custom-3</i>	<i>reserved</i>

Table 72 shows a map of the major opcodes for RVG. Opcodes marked as *reserved* should be avoided for custom instruction-set extensions as they might be used by future standard extensions. Major opcodes marked as *custom-0* through *custom-3* will be avoided by future standard extensions and are recommended for use by custom instruction-set extensions within the base 32-bit instruction format.

We believe RV32G and RV64G provide simple but complete instruction sets for a broad range of general-purpose computing. The optional compressed instruction set described in Chapter 28 can be added (forming RV32GC and RV64GC) to improve performance, code size, and energy efficiency, though with some additional hardware complexity.

As we move beyond IMAFDC into further instruction-set extensions, the added instructions tend to be more domain-specific and only provide benefits to a restricted class of applications, e.g., for multimedia or security. Unlike most commercial ISAs, the RISC-V ISA design clearly separates the base ISA and broadly applicable standard extensions from these more specialized additions.

31	27	26	25	24	20	19	15	14	12	11		7	6		0	
funct7				rs2		rs1		funct3		rd			opcode		R-type	
imm[11:0]						rs1		funct3		rd			opcode		I-type	
imm[11:5]				rs2		rs1		funct3		imm[4:0]			opcode		S-type	
imm[12 10:5]				rs2		rs1		funct3		imm[4:1 11]			opcode		B-type	
imm[31:12]										rd			opcode		U-type	
imm[20 10:1 11 19:12]										rd			opcode		J-type	

RV32I Base Instruction Set						
imm[31:12]				rd		0110111 LUI
imm[31:12]				rd		0010111 AUIPC
imm[20 10:1 11 19:12]				rd		1101111 JAL
imm[11:0]			rs1	000	rd	1100111 JALR
imm[12 10:5]	rs2	rs1	000	imm[4:1 11]	1100011	BEQ
imm[12 10:5]	rs2	rs1	001	imm[4:1 11]	1100011	BNE
imm[12 10:5]	rs2	rs1	100	imm[4:1 11]	1100011	BLT
imm[12 10:5]	rs2	rs1	101	imm[4:1 11]	1100011	BGE
imm[12 10:5]	rs2	rs1	110	imm[4:1 11]	1100011	BLTU
imm[12 10:5]	rs2	rs1	111	imm[4:1 11]	1100011	BGEU
imm[11:0]			rs1	000	rd	0000011 LB
imm[11:0]			rs1	001	rd	0000011 LH
imm[11:0]			rs1	010	rd	0000011 LW
imm[11:0]			rs1	100	rd	0000011 LBU
imm[11:0]			rs1	101	rd	0000011 LHU
imm[11:5]	rs2	rs1	000	imm[4:0]	0100011	SB
imm[11:5]	rs2	rs1	001	imm[4:0]	0100011	SH
imm[11:5]	rs2	rs1	010	imm[4:0]	0100011	SW
imm[11:0]			rs1	000	rd	0010011 ADDI
imm[11:0]			rs1	010	rd	0010011 SLTI
imm[11:0]			rs1	011	rd	0010011 SLTIU
imm[11:0]			rs1	100	rd	0010011 XORI
imm[11:0]			rs1	110	rd	0010011 ORI
imm[11:0]			rs1	111	rd	0010011 ANDI
0000000	shamt	rs1	001	rd	0010011	SLLI
0000000	shamt	rs1	101	rd	0010011	SRLI
0100000	shamt	rs1	101	rd	0010011	SRAI
0000000	rs2	rs1	000	rd	0110011	ADD
0100000	rs2	rs1	000	rd	0110011	SUB
0000000	rs2	rs1	001	rd	0110011	SLL

0000000		rs2	rs1	010	rd	0110011	SLT
0000000		rs2	rs1	011	rd	0110011	SLTU
0000000		rs2	rs1	100	rd	0110011	XOR
0000000		rs2	rs1	101	rd	0110011	SRL
0100000		rs2	rs1	101	rd	0110011	SRA
0000000		rs2	rs1	110	rd	0110011	OR
0000000		rs2	rs1	111	rd	0110011	AND
fm	pred	succ	rs1	000	rd	0001111	FENCE
1000	0011	0011	00000	000	00000	0001111	FENCE.TSO
0000	0001	0000	00000	000	00000	0001111	PAUSE
0000000000000			00000	000	00000	1110011	ECALL
0000000000001			00000	000	00000	1110011	EBREAK

31	27	26	25	24	20	19	15	14	12	11	7	6	0	
funct7				rs2		rs1		funct3		rd		opcode		R-type
imm[11:0]						rs1		funct3		rd		opcode		I-type
imm[11:5]				rs2		rs1		funct3		imm[4:0]		opcode		S-type

RV64I Base Instruction Set (in addition to RV32I)

imm[11:0]				rs1		110		rd		0000011		LWU	
imm[11:0]				rs1		011		rd		0000011		LD	
imm[11:5]			rs2		rs1		011		imm[4:0]		0100011		SD
000000			shamt		rs1		001		rd		0010011		SLLI
000000			shamt		rs1		101		rd		0010011		SRLI
010000			shamt		rs1		101		rd		0010011		SRAI
imm[11:0]				rs1		000		rd		0011011		ADDIW	
0000000			shamt		rs1		001		rd		0011011		SLLIW
0000000			shamt		rs1		101		rd		0011011		SRLIW
0100000			shamt		rs1		101		rd		0011011		SRAIW
0000000			rs2		rs1		000		rd		0111011		ADDW
0100000			rs2		rs1		000		rd		0111011		SUBW
0000000			rs2		rs1		001		rd		0111011		SLLW
0000000			rs2		rs1		101		rd		0111011		SRLW
0100000			rs2		rs1		101		rd		0111011		SRAW

RV32/RV64 Zifencei Standard Extension

imm[11:0]				rs1		001		rd		0001111		FENCE.I	
-----------	--	--	--	-----	--	-----	--	----	--	---------	--	---------	--

RV32/RV64 Zicsr Standard Extension

csr				rs1		001		rd		1110011		CSRRW	
csr				rs1		010		rd		1110011		CSRRS	
csr				rs1		011		rd		1110011		CSRRC	
csr				uimm		101		rd		1110011		CSRRWI	
csr				uimm		110		rd		1110011		CSRRSI	
csr				uimm		111		rd		1110011		CSRRCI	

RV32M Standard Extension

0000001			rs2		rs1		000		rd		0110011		MUL
0000001			rs2		rs1		001		rd		0110011		MULH
0000001			rs2		rs1		010		rd		0110011		MULHSU
0000001			rs2		rs1		011		rd		0110011		MULHU
0000001			rs2		rs1		100		rd		0110011		DIV
0000001			rs2		rs1		101		rd		0110011		DIVU

0000001	rs2	rs1	110	rd	0110011	REM
0000001	rs2	rs1	111	rd	0110011	REMU

RV64M Standard Extension (in addition to RV32M)						
0000001	rs2	rs1	000	rd	0111011	MULW
0000001	rs2	rs1	100	rd	0111011	DIVW
0000001	rs2	rs1	101	rd	0111011	DIVUW
0000001	rs2	rs1	110	rd	0111011	REMW
0000001	rs2	rs1	111	rd	0111011	REMUW

31	27	26	25	24	20	19	15	14	12	11	7	6	0	
funct7				rs2		rs1		funct3		rd		opcode		R-type

RV32A Standard Extension								
00010	aq	rL	00000	rs1	010	rd	0101111	LR.W
00011	aq	rL	rs2	rs1	010	rd	0101111	SC.W
00001	aq	rL	rs2	rs1	010	rd	0101111	AMOSWAP.W
00000	aq	rL	rs2	rs1	010	rd	0101111	AMOADD.W
00100	aq	rL	rs2	rs1	010	rd	0101111	AMOXOR.W
01100	aq	rL	rs2	rs1	010	rd	0101111	AMOAND.W
01000	aq	rL	rs2	rs1	010	rd	0101111	AMOOR.W
10000	aq	rL	rs2	rs1	010	rd	0101111	AMOMIN.W
10100	aq	rL	rs2	rs1	010	rd	0101111	AMOMAX.W
11000	aq	rL	rs2	rs1	010	rd	0101111	AMOMINU.W
11100	aq	rL	rs2	rs1	010	rd	0101111	AMOMAXU.W

RV64A Standard Extension (in addition to RV32A)								
00010	aq	rL	00000	rs1	011	rd	0101111	LR.D
00011	aq	rL	rs2	rs1	011	rd	0101111	SC.D
00001	aq	rL	rs2	rs1	011	rd	0101111	AMOSWAP.D
00000	aq	rL	rs2	rs1	011	rd	0101111	AMOADD.D
00100	aq	rL	rs2	rs1	011	rd	0101111	AMOXOR.D
01100	aq	rL	rs2	rs1	011	rd	0101111	AMOAND.D
01000	aq	rL	rs2	rs1	011	rd	0101111	AMOOR.D
10000	aq	rL	rs2	rs1	011	rd	0101111	AMOMIN.D
10100	aq	rL	rs2	rs1	011	rd	0101111	AMOMAX.D
11000	aq	rL	rs2	rs1	011	rd	0101111	AMOMINU.D
11100	aq	rL	rs2	rs1	011	rd	0101111	AMOMAXU.D

31	27	26	25	24	20	19	15	14	12	11	7	6	0	
funct7				rs2		rs1		funct3		rd		opcode		R-type
rs3		funct2		rs2		rs1		funct3		rd		opcode		R4-type
imm[11:0]						rs1		funct3		rd		opcode		I-type
imm[11:5]				rs2		rs1		funct3		imm[4:0]		opcode		S-type

RV32F Standard Extension							
imm[11:0]			rs1	010	rd	0000111	FLW
imm[11:5]		rs2	rs1	010	imm[4:0]	0100111	FSW
rs3	00	rs2	rs1	rm	rd	1000011	FMADD.S
rs3	00	rs2	rs1	rm	rd	1000111	FMSUB.S
rs3	00	rs2	rs1	rm	rd	1001011	FNMSUB.S
rs3	00	rs2	rs1	rm	rd	1001111	FNMADD.S
0000000		rs2	rs1	rm	rd	1010011	FADD.S
0000100		rs2	rs1	rm	rd	1010011	FSUB.S
0001000		rs2	rs1	rm	rd	1010011	FMUL.S
0001100		rs2	rs1	rm	rd	1010011	FDIV.S
0101100		00000	rs1	rm	rd	1010011	FSQRT.S
0010000		rs2	rs1	000	rd	1010011	FSGNJ.S
0010000		rs2	rs1	001	rd	1010011	FSGNJN.S
0010000		rs2	rs1	010	rd	1010011	FSGNJX.S
0010100		rs2	rs1	000	rd	1010011	FMIN.S
0010100		rs2	rs1	001	rd	1010011	FMAX.S
1100000		00000	rs1	rm	rd	1010011	FCVT.W.S
1100000		00001	rs1	rm	rd	1010011	FCVT.WU.S
1110000		00000	rs1	000	rd	1010011	FMV.X.W
1010000		rs2	rs1	010	rd	1010011	FEQ.S
1010000		rs2	rs1	001	rd	1010011	FLT.S
1010000		rs2	rs1	000	rd	1010011	FLE.S
1110000		00000	rs1	001	rd	1010011	FCLASS.S
1101000		00000	rs1	rm	rd	1010011	FCVT.S.W
1101000		00001	rs1	rm	rd	1010011	FCVT.S.WU
1111000		00000	rs1	000	rd	1010011	FMV.W.X

RV64F Standard Extension (in addition to RV32F)														
1100000			00010	rs1	rm	rd		1010011		FCVT.L.S				
1100000			00011	rs1	rm	rd		1010011		FCVT.LU.S				
1101000			00010	rs1	rm	rd		1010011		FCVT.S.L				
1101000			00011	rs1	rm	rd		1010011		FCVT.S.LU				

31	27	26	25	24	20	19	15	14	12	11	7	6	0	
funct7				rs2		rs1		funct3		rd		opcode		R-type
rs3		funct2		rs2		rs1		funct3		rd		opcode		R4-type
imm[11:0]						rs1		funct3		rd		opcode		I-type
imm[11:5]				rs2		rs1		funct3		imm[4:0]		opcode		S-type

RV32D Standard Extension							
imm[11:0]			rs1	011	rd	0000111	FLD
imm[11:5]		rs2	rs1	011	imm[4:0]	0100111	FSD
rs3	01	rs2	rs1	rm	rd	1000011	FMADD.D
rs3	01	rs2	rs1	rm	rd	1000111	FMSUB.D
rs3	01	rs2	rs1	rm	rd	1001011	FNMSUB.D
rs3	01	rs2	rs1	rm	rd	1001111	FNMADD.D
0000001		rs2	rs1	rm	rd	1010011	FADD.D
0000101		rs2	rs1	rm	rd	1010011	FSUB.D
0001001		rs2	rs1	rm	rd	1010011	FMUL.D
0001101		rs2	rs1	rm	rd	1010011	FDIV.D
0101101		00000	rs1	rm	rd	1010011	FSQRT.D
0010001		rs2	rs1	000	rd	1010011	FSGNJ.D
0010001		rs2	rs1	001	rd	1010011	FSGNJN.D
0010001		rs2	rs1	010	rd	1010011	FSGNJX.D
0010101		rs2	rs1	000	rd	1010011	FMIN.D
0010101		rs2	rs1	001	rd	1010011	FMAX.D
0100000		00001	rs1	rm	rd	1010011	FCVT.S.D
0100001		00000	rs1	rm	rd	1010011	FCVT.D.S
1010001		rs2	rs1	010	rd	1010011	FEQ.D
1010001		rs2	rs1	001	rd	1010011	FLT.D
1010001		rs2	rs1	000	rd	1010011	FLE.D
1110001		00000	rs1	001	rd	1010011	FCLASS.D
1100001		00000	rs1	rm	rd	1010011	FCVT.W.D
1100001		00001	rs1	rm	rd	1010011	FCVT.WU.D
1101001		00000	rs1	rm	rd	1010011	FCVT.D.W
1101001		00001	rs1	rm	rd	1010011	FCVT.D.WU

RV64D Standard Extension (in addition to RV32D)													
1100001				00010	rs1	rm	rd	1010011		FCVT.L.D			
1100001				00011	rs1	rm	rd	1010011		FCVT.LU.D			
1110001				00000	rs1	000	rd	1010011		FMV.X.D			
1101001				00010	rs1	rm	rd	1010011		FCVT.D.L			
1101001				00011	rs1	rm	rd	1010011		FCVT.D.LU			

1111001	00000	rs1	000	rd	1010011	FMV.D.X
---------	-------	-----	-----	----	---------	---------